Copter Experiment Report

Nicholas Diamond

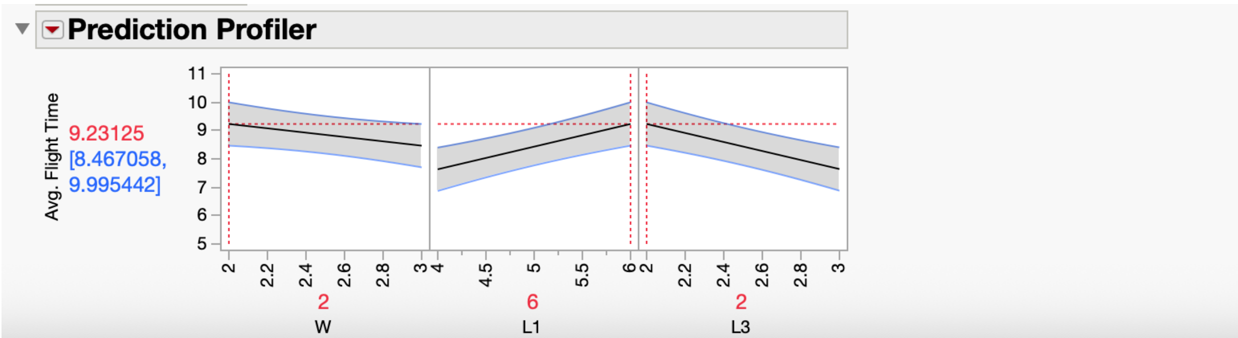
Managerial report

We conducted an experiment to determine how different design factors, such as blade length, blade width, and body length, affect flight performance and find an optimal design.

The variable we used to get a response in this experiment is the paper copter's flight time, measured in seconds. Flight time is the total duration from when the copter is released from a fixed height until it contacts the ground. This measurement serves as a direct indicator of the aerodynamic performance of each copter design. In this experiment, three factors were varied to examine their effects on the flight time of a paper copter. The first factor, blade width (W), refers to the horizontal span of the rotor blades. Changing the width alters the surface area exposed to air resistance. The second factor, blade length (L1), is the vertical length of each blade. The third factor, body length (L3), represents the vertical length of the copter’s central body. These factors were systematically manipulated to evaluate their main effects and any interaction effects on the response variable, flight time. The distance between the blades and the body (L2) was constant at 1 inch. Nicholas Diamond, Mac Griffen, and Nolan Conoly Brown are the conductors of this experiment.

The experiment utilized a full factorial design with three factors: blade width, blade length, and body length. Each factor was tested at two levels (high and low), resulting in eight unique design combinations. For each combination, two copters were constructed to account for variations in building (2 replications), and each copter was flown twice (2 repetitions). This resulted in a total of four observations per design combination, culminating in 32 trials overall. Flight time served as the performance metric. Each copter was dropped from a consistent height, from a second-story balcony, and the drop time was measured using a stopwatch. A “3, 2, 1” countdown ensured consistent timing for the release, and flight time was recorded from the moment of release until the copter contacted the ground.

A single team member constructed all copters to ensure consistency in both design and assembly. Each copter was cut from a larger sheet of paper using a standardized template. A one-inch section separated the blades (L1) from the body (L3), and a one-inch base was maintained for balance. A single piece of tape was used for each copter, applied in the exact location and with the same fold style across all builds. We had a practice run for each copter to ensure they functioned properly and minimize errors. This controlled approach minimized variability caused by the construction and ensured that any performance differences were mainly attributed to the experimental factors.



The experiment demonstrated that blade width, blade length, and body length statistically significantly affected flight time. The best-performing design combined a narrow blade width (2 inches), a long blade length (6 inches), and a short body length (2 inches). The most influential factor was blade length and the length of the body; longer blades consistently led to longer flight times. Shortening the body length also improved performance, likely by reducing the copter's weight and improving balance during descent.

Although the effect of blade width was negligible, narrower blades still resulted in slightly longer and more stable flights. These trends were consistent across the data and clearly illustrated in the profiler, where each factor showed a measurable and reliable impact on performance. The experiment confirms that specific design choices can significantly improve the aerodynamic efficiency of a paper copter. Although the results of this experiment were statistically significant and offered clear guidance for optimizing the design of paper copters, a valuable next step would be to investigate more extreme values for blade length (L1) and body length (L3). Specifically, testing longer blades alongside shorter bodies could help determine whether the observed trends continue beyond the current levels or if they eventually reach a point of diminishing returns. By expanding the range of these factors, we could gain deeper insights into the limits of design improvements, potentially leading to even greater increases in flight time.

Technical Report

Objectives

The experimental objective is to determine how different design factors, such as blade length, blade width, and body length, affect flight performance and find an optimal design.

Response Variable (Y) and Experimental Factors (X)

The response variable in this experiment is the flight time of the paper copter, measured in seconds. Flight time is the total duration from when the copter is released from a fixed height until it contacts the ground. This measurement serves as a direct indicator of the aerodynamic performance of each copter design. By analyzing variations in flight time, we can evaluate how different design factors influence the descent rate and overall effectiveness of the copter. In this experiment, three factors were varied to examine their effects on the flight time of a paper copter. The first factor, blade width (W), refers to the horizontal span of the rotor blades, which was either 2 or 3 inches. Changing the width alters the surface area exposed to air resistance. The second factor, blade length (L1), is the vertical length of each blade, which was between 4 and 6 inches. The third factor, body length (L3), represents the vertical length of the copter's central body, which was between 2 and 3 inches. This dimension affects the center of mass and rotational dynamics. These factors were systematically manipulated to evaluate their main effects and any interaction effects on the response variable, flight time. The distance between the blades and the body (L2) was constant at 1 inch.

Design and Measurement Method of the Experiment

This experiment utilized a full factorial design to investigate the impact of three design factors on the flight time of paper copters: blade width (W), blade length (L1), and body length (L3). Each factor was tested at two levels: blade width at 2 inches and 3 inches, blade length at 4 inches and 6 inches, and body length at 2 inches and 3 inches. These combinations resulted in eight unique copter designs. For each design, two separate copters were constructed (replications), and each copter was flown twice (repetitions), leading to four trials per design and 32 data points overall. The order of the runs was randomized to minimize bias.

All copters were built by a single team member using a consistent construction process. Each design was cut from a larger sheet of paper using a standardized template that included a one-inch space between the blades and the body and a one-inch base at the bottom. A single piece of tape was applied uniformly across all designs in the same position and folding style to ensure consistency in assembly. The copters were handled carefully between flights to avoid deformation.

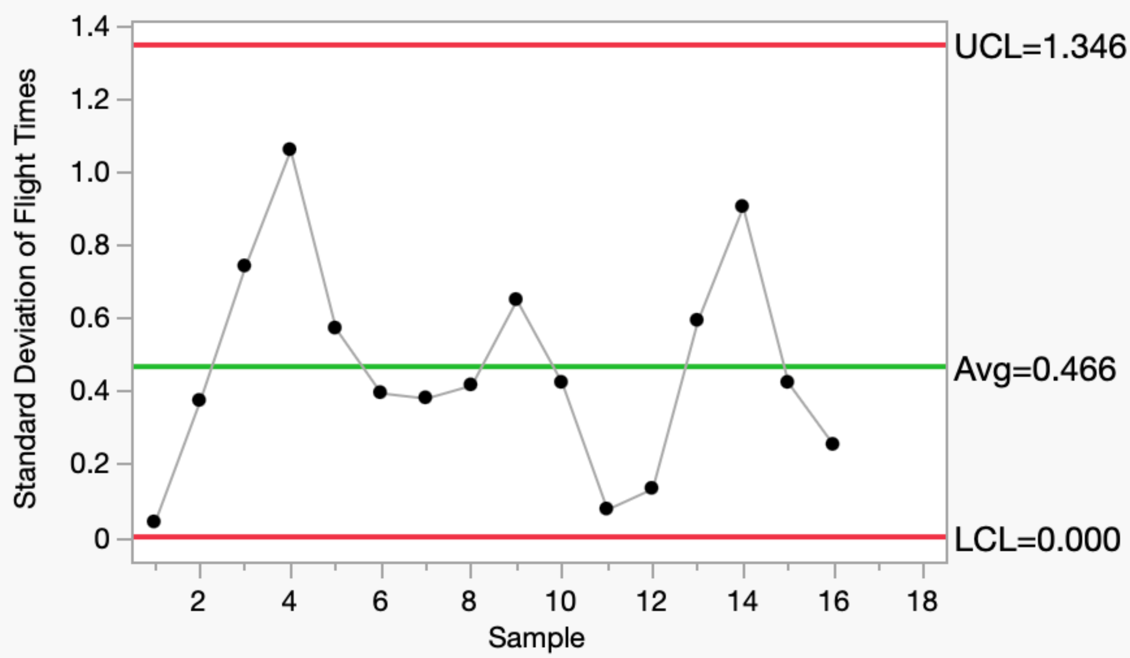
Flight time was measured from the moment of release until ground contact. Each copter was dropped from the same second-story balcony to ensure a consistent descent distance. A standardized “3, 2, 1” countdown was used to synchronize the release and timing: one team member released the copter while another simultaneously started a stopwatch. All flights were timed using the same stopwatch for every trial, and the readings were recorded immediately. To minimize errors, environmental conditions were kept as consistent as possible. Flights were conducted indoors to eliminate wind interference, and the roles of the stopwatch operator and the person releasing the copter remained the same throughout the experiment. While some uncontrolled variations, such as differences in hand release position or reaction timing, may have occurred, the controlled design and repetition of the trials helped ensure reliable and valid measurements.

To better understand the sources of variation in the experiment, we categorized variables into two groups. Bucket One included controlled factor: using one piece of tape per copter, a standardized “3, 2, 1” countdown, consistent roles for team members, releasing each copter from the same second-story height, maintaining a one-inch space between the blades and body, using a one-inch base on every copter, and following construction instructions. These controls ensured that any differences in flight time were due to the experimental factors being studied. Bucket Two encompassed uncontrolled sources of variation, such as wind, nearby people, stopwatch timing inconsistencies, creased paper upon landing, natural flight-to-flight variations, and minor cutting or folding inconsistencies. While these factors could not be eliminated, recognizing their influence helps understand some data variation.

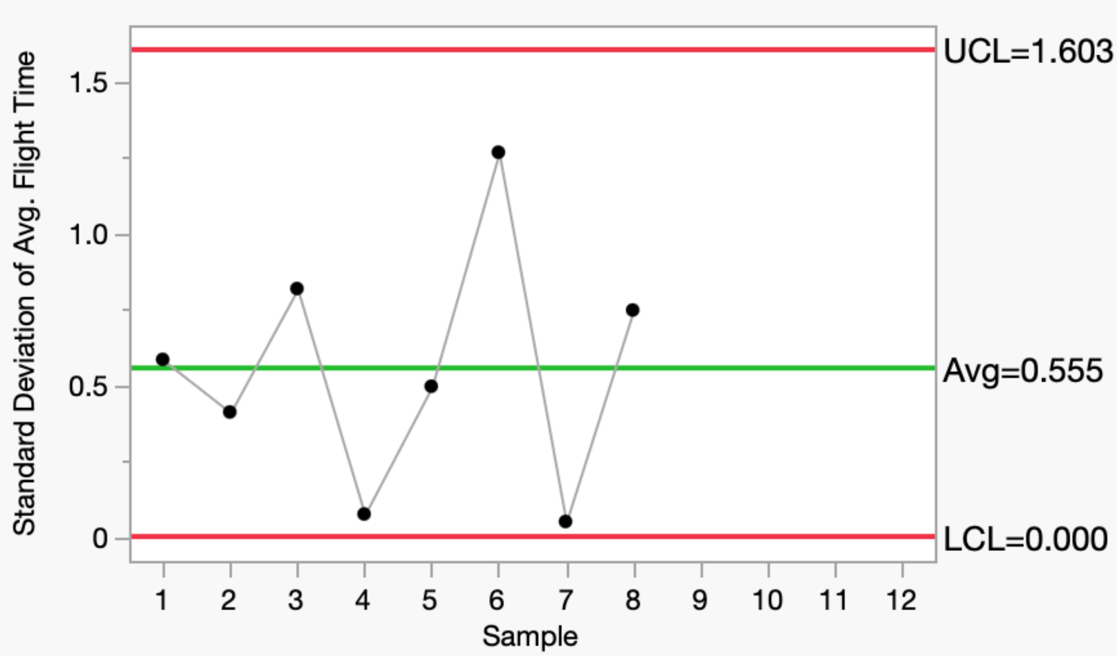
Experiment Results

	Team	Pattern	Copter Number (Run Order)	W	L1	L3	Flight 1	Flight 2	Avg. Flight Time	Width	Length
1	09	−−+	1	2	4	3	6	6.06	6.03	4	8
2	09	++−	2	3	6	2	8.53	8	8.265	6	9
3	09	+++	3	3	6	3	5.79	6.84	6.315	6	10
4	09	−+−	4	2	6	2	11.4	9.9	10.65	4	9
5	09	+−−	5	3	4	2	6.69	5.88	6.285	6	7
6	09	−++	6	2	6	3	7.81	7.25	7.53	4	10
7	09	−−−	7	2	4	2	7.81	8.35	8.08	4	7
8	09	+−+	8	3	4	3	5	5.59	5.295	6	8
9	other	+−−	1	3	4	2	5.93	6.85	6.39	6	7
10	other	−++	2	2	6	3	6.53	7.13	6.83	4	10
11	other	++−	3	3	6	2	8.9	8.79	8.845	6	9
12	other	−−−	4	2	4	2	6.93	7.12	7.025	4	7
13	other	−+−	5	2	6	2	9.28	8.44	8.86	4	9
14	other	+++	6	3	6	3	6.5	7.78	7.14	6	10
15	other	−−+	7	2	4	3	5.8	6.4	6.1	4	8
16	other	+−+	8	3	4	3	6.63	6.27	6.45	6	8

S of Flight Times



S of Avg. Flight Time



Two S-charts were created to assess the stability of variation in the experiment. The first chart analyzes the standard deviation of repeated flight times for each individual copter, based on 16 data points. The second chart analyzes the standard deviation of average flight times across different builds of the same design, using 8 data points. These charts are designed to detect any special cause variation that could indicate inconsistencies in how the copters were flown, built, or measured.

In the repetition S-chart, all 16 data points fall within the control limits, Upper Control Limit (UCL) = 1.346, Lower Control Limit (LCL) = 0.000), with an average standard deviation of 0.466 seconds. Although some variation is present, particularly in Samples 4 and 5, no point exceeds the control limits, indicating that the variation between repeated flights of the same copter was stable and consistent. The replication S-chart displays the standard deviation of average flight times across the two copters built for each design. All eight samples fall within the control limits (UCL = 1.603, LCL = 0.000), and the average standard deviation is 0.555 seconds. No unusual variation is observed, confirming that differences between replicated builds of the same design remain within expected limits.

If points had fallen outside the control limits on either chart, it would have signaled special cause variation. In such cases, the corresponding copter would have been rebuilt according to the original specifications, re-flown, and the data would have been replaced to maintain the experiment's integrity. Since all data points are within control limits, we can conclude that the experiment was conducted using consistent methods, and the resulting data is reliable for further analysis.

Summary of Fit

RSquare	0.793178
RSquare Adj	0.741473
Root Mean Square Error	0.701475
Mean of Response	7.255625
Observations (or Sum Wgts)	16

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	22.645431	7.54848	15.3403
Error	12	5.904812	0.49207	Prob > F
C. Total	15	28.550244		0.0002*

Lack Of Fit

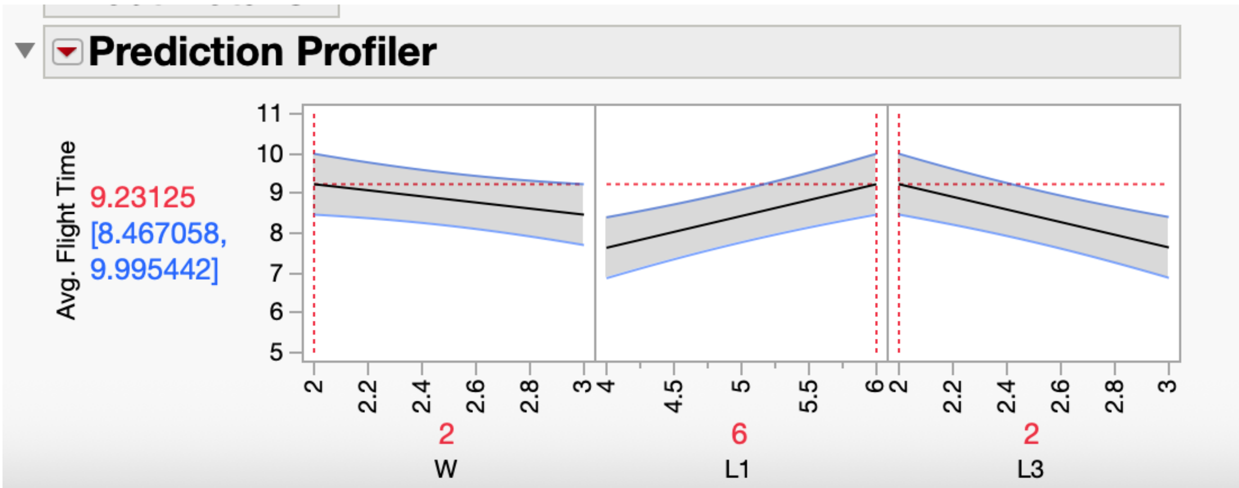
Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	4	2.3177625	0.579441	1.2923
Pure Error	8	3.5870500	0.448381	Prob > F
Total Error	12	5.9048125		0.3500
				Max RSq
				0.8744

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	Uncoded Estimate
Intercept	7.255625	0.175369	41.37	<.0001*	9.14625
W(2,3)	-0.3825	0.175369	-2.18	0.0498*	-0.765
L1(4,6)	0.79875	0.175369	4.55	0.0007*	0.79875
L3(2,3)	-0.794375	0.175369	-4.53	0.0007*	-1.58875

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
W(2,3)	1	1	2.340900	4.7573	0.0498*
L1(4,6)	1	1	10.208025	20.7452	0.0007*
L3(2,3)	1	1	10.096506	20.5185	0.0007*



After conducting our initial analysis and confirming statistical control using S-charts, we moved forward with a full factorial regression analysis to explore how design factors influenced flight time. Using JMP software, we modeled flight time as a function of blade width (W), blade length (L1), and body length (L3), with each factor tested at two levels. The model produced an R-squared value of 0.79, indicating that the design variables explained a significant portion of the variability in flight time. The lack-of-fit test revealed no significant errors beyond random variation, supporting the validity of our model. The experiment demonstrated that blade width, blade length, and body length all had a statistically significant impact on flight time. The optimal design was a combination of a narrow blade width (2 inches), a long blade length (6 inches), and a short body length (2 inches). Among the three factors, blade length and body length were the most influential. Longer blade lengths consistently resulted in longer flight times, while shorter body lengths improved performance, likely by reducing weight and enhancing aerodynamic stability. Although the effect of blade width was less pronounced, narrower blades still contributed to slightly longer and more stable descents. These trends are clearly illustrated in the prediction profiler, where each factor shows a distinct and reliable slope, reinforcing their individual impact on average flight time. No significant interaction effects were observed, indicating that the influence of each factor operated independently of the others. This consistency in the data confirms that specific design choices, particularly increasing blade length and decreasing body length, can significantly enhance the aerodynamic efficiency of a paper copter. While these results are statistically significant and strongly support design optimization, the current factor levels tested were relatively moderate

Follow Up

Although this experiment yielded statistically significant results and provided clear guidance for optimizing the design of paper copters, a valuable next step would be to explore more extreme values for blade length (L1) and body length (L3). Specifically, testing longer blades alongside shorter body lengths could help determine whether the trends observed in this study continue beyond the tested range or if they plateau. Expanding the range of these factors would allow for a more thorough investigation of aerodynamic limits and could lead to even greater improvements in flight time. This follow-up experiment would also enhance the generalizability of our conclusions and help identify the optimal boundaries of design performance.